

# Watch-DevKit-HW

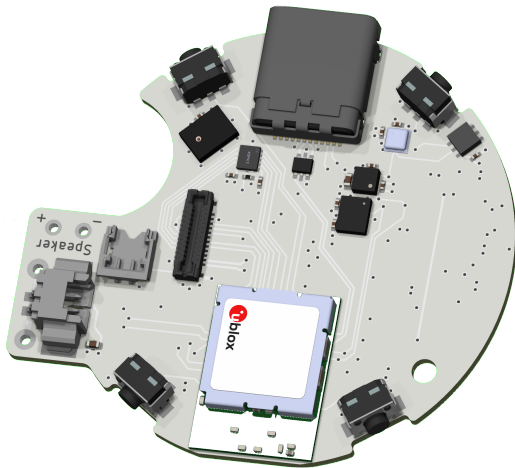
## Variant: PRELIMINARY

2026-09-04

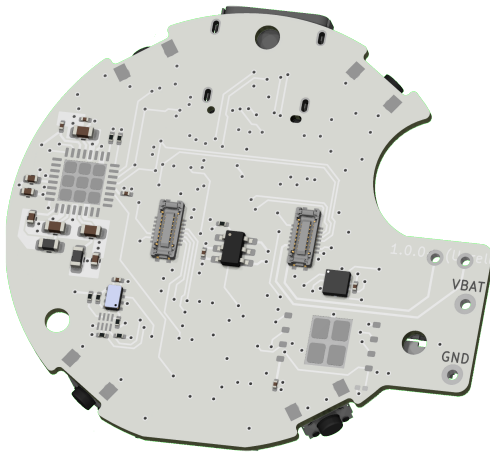
Rev 1.0.0+ (Unreleased)

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TOP VIEW



BOTTOM VIEW



### NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 04-Sep-2026

### DESIGN CONSIDERATIONS

DESIGN NOTE:  
Example text for  
informational design  
notes.

DESIGN NOTE:  
Example text for  
debug notes.

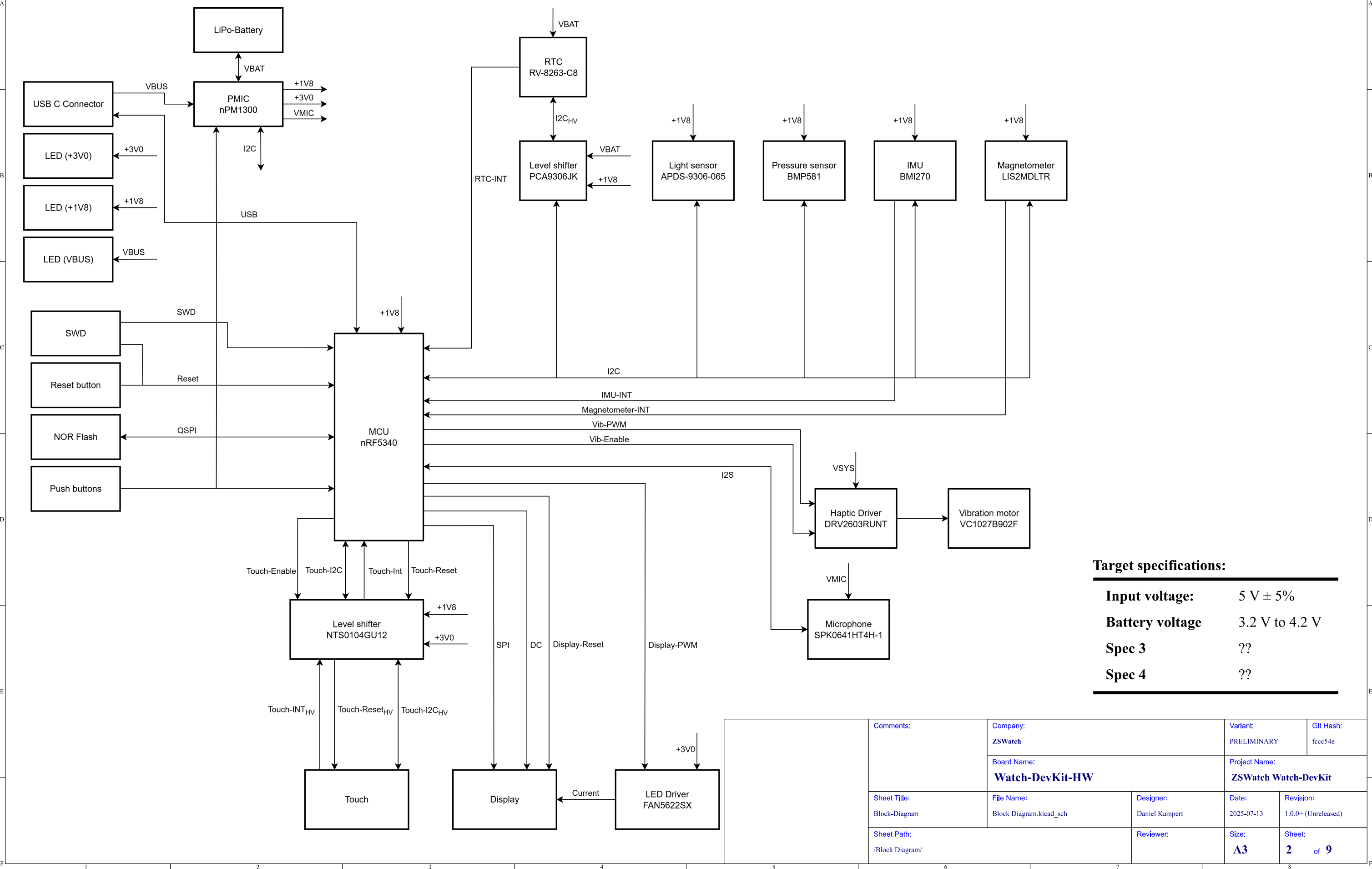
DESIGN NOTE:  
Example text for  
cautionary design  
notes.

DESIGN NOTE:  
Example text for  
critical design  
notes.

LAYOUT NOTE:  
Example text for  
critical layout  
guidelines.

|  |                                |                                       |                                       |                                  |
|--|--------------------------------|---------------------------------------|---------------------------------------|----------------------------------|
|  | Comments:                      | Company:<br>ZSWatch                   | Variant:<br>PRELIMINARY               | Git Hash:<br>fccc54e             |
|  | Board Name:<br>Watch-DevKit-HW |                                       | Project Name:<br>ZSWatch Watch-DevKit |                                  |
|  | Sheet Title:<br>ZSWatch-Watch  | File Name:<br>ZSWatch-Watch.kicad_sch | Designer:<br>Daniel Kampert           | Date:<br>2025-07-13              |
|  | Sheet Path:<br>/               |                                       | Reviewer:                             | Revision:<br>1.0.0+ (Unreleased) |
|  |                                |                                       | Size:<br>A3                           | Sheet:<br>1 of 9                 |

[2] Block-Diagram



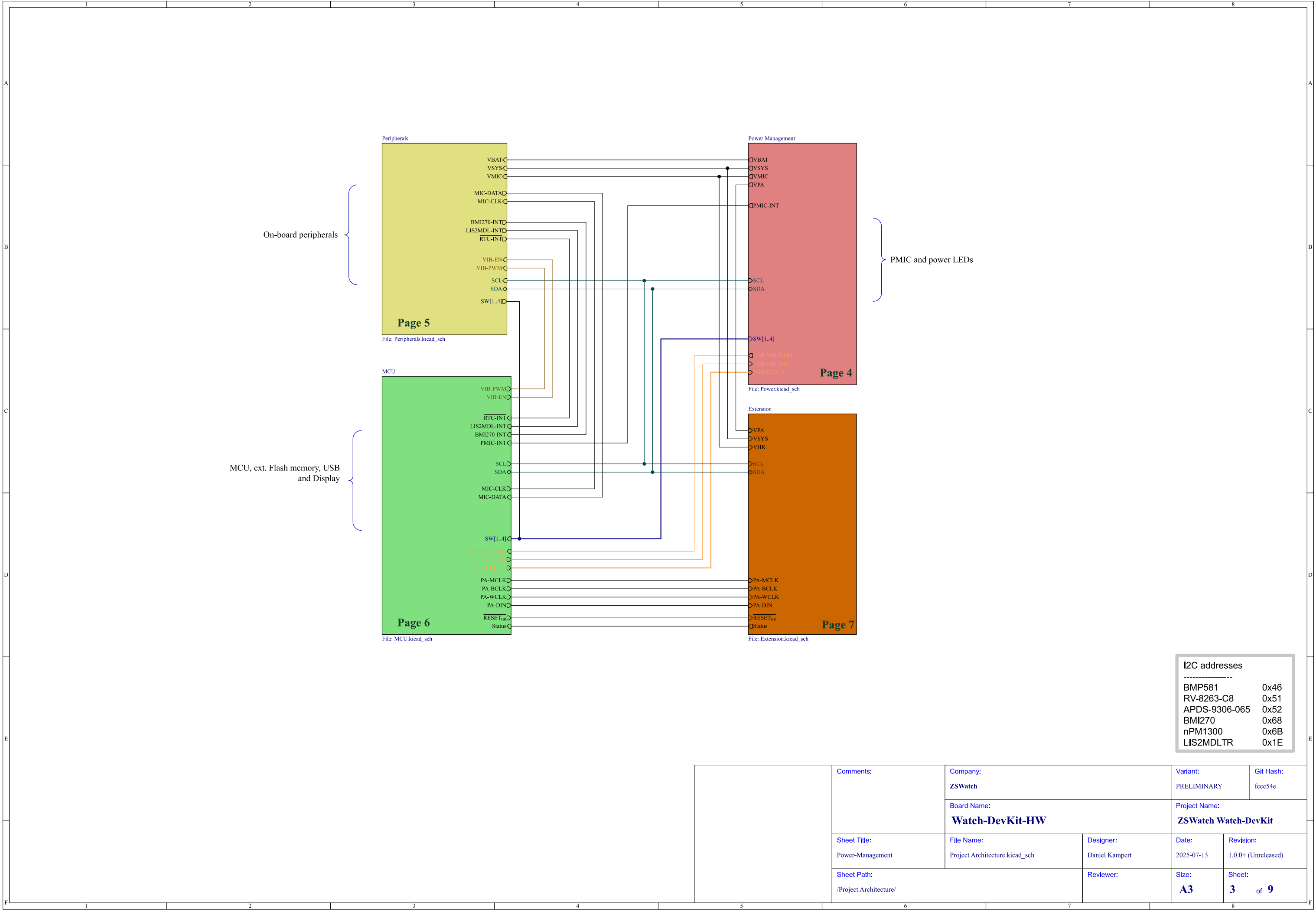
| Target specifications: |                |  |  |  |
|------------------------|----------------|--|--|--|
| Input voltage:         | 5 V ± 5%       |  |  |  |
| Battery voltage        | 3.2 V to 4.2 V |  |  |  |
| Spec 3                 | ??             |  |  |  |
| Spec 4                 | ??             |  |  |  |

|               |                                |                                       |                             |                                              |                                  |
|---------------|--------------------------------|---------------------------------------|-----------------------------|----------------------------------------------|----------------------------------|
| Block-Diagram | Comments:                      | Company:<br><b>ZSWatch</b>            |                             | Variant:<br>PRELIMINARY                      | Git Hash:<br>fcc54e              |
|               | Sheet Title:                   | Board Name:<br><b>Watch-DevKit-HW</b> |                             | Project Name:<br><b>ZSWatch Watch-DevKit</b> |                                  |
|               | Sheet Path:<br>/Block Diagram/ | File Name:<br>Block Diagram.kicad_sch | Designer:<br>Daniel Kampert | Date:<br>2025-07-13                          | Revision:<br>1.0.0+ (Unreleased) |
|               |                                |                                       | Reviewer:                   | Size:<br><b>A3</b>                           | Sheet:<br><b>2</b> of <b>9</b>   |

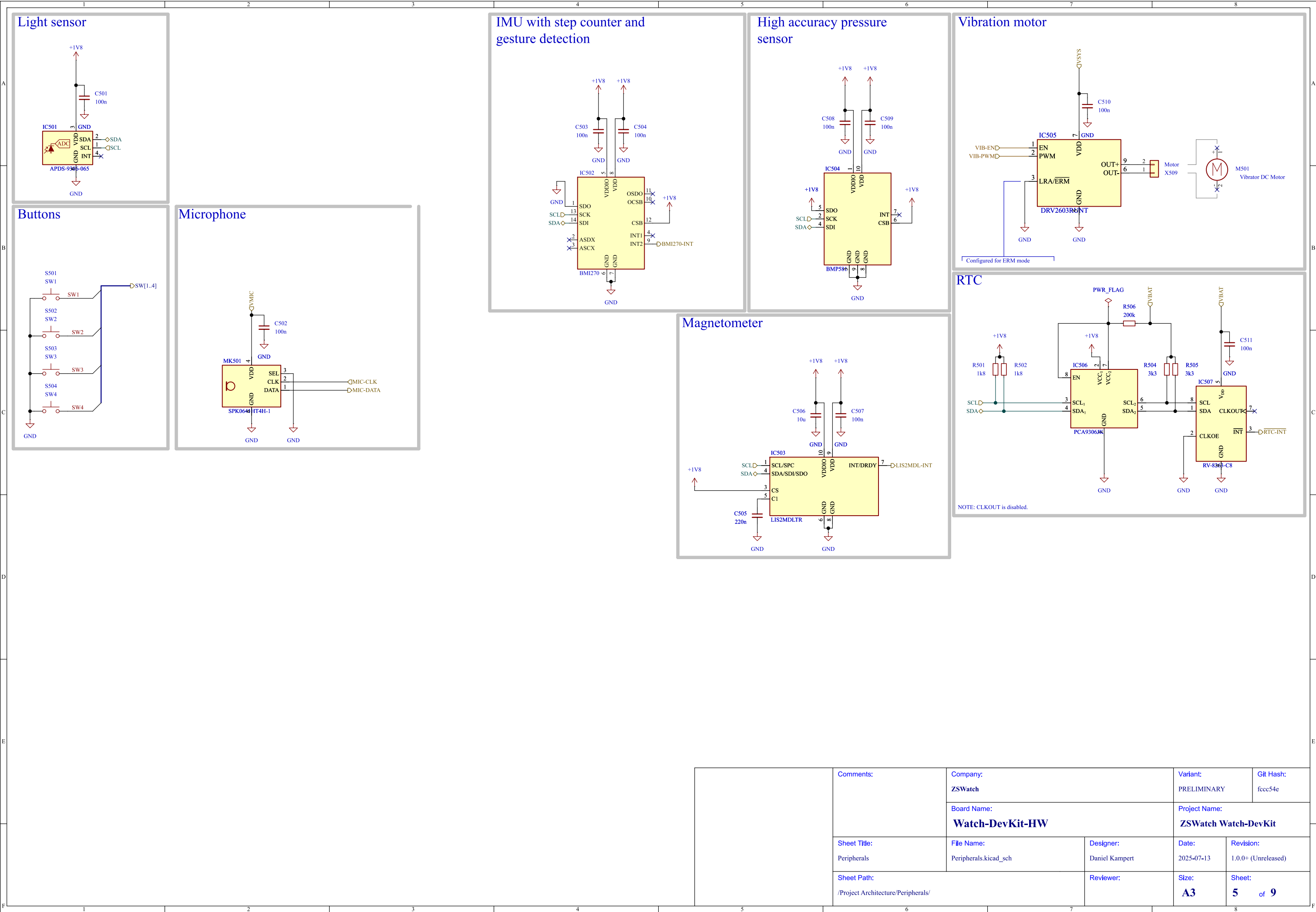
|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| A |   |   |   |   |   |   |   |   |
| B |   |   |   |   |   |   |   |   |
| C |   |   |   |   |   |   |   |   |
| D |   |   |   |   |   |   |   |   |
| E |   |   |   |   |   |   |   |   |
| F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

| Target specifications: |           |
|------------------------|-----------|
| Input voltage:         | ?? - ?? V |
| Spec 2                 | ??        |
| Spec 3                 | ??        |
| Spec 4                 | ??        |

|  |                                   |                                          |                             |                                       |                                  |
|--|-----------------------------------|------------------------------------------|-----------------------------|---------------------------------------|----------------------------------|
|  | Comments:                         | Company:<br>ZSWatch                      |                             | Variant:<br>PRELIMINARY               | Git Hash:<br>fccc54e             |
|  |                                   | Board Name:<br>Watch-DevKit-HW           |                             | Project Name:<br>ZSWatch Watch-DevKit |                                  |
|  | Sheet Title:                      | File Name:<br>Power - Topology.kicad_sch | Designer:<br>Daniel Kampert | Date:<br>2025-07-13                   | Revision:<br>1.0.0+ (Unreleased) |
|  | Sheet Path:<br>/Power - Topology/ |                                          | Reviewer:                   | Size:<br>A3                           | Sheet:<br>2 of 9                 |

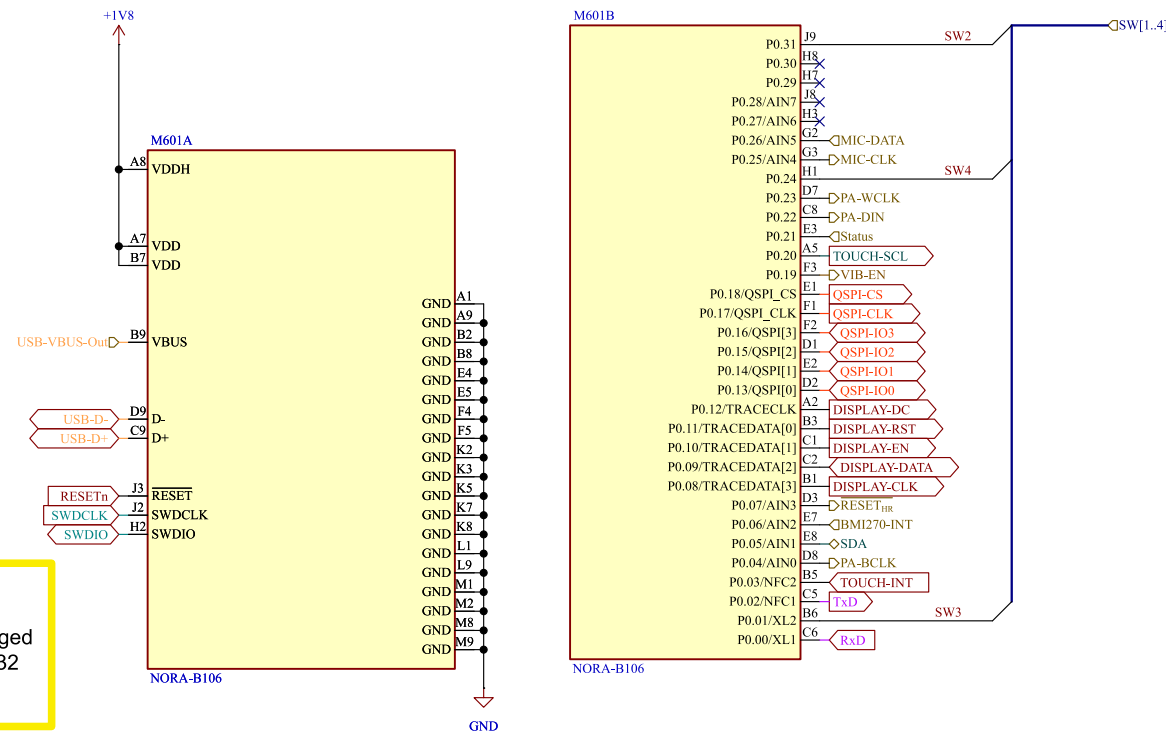






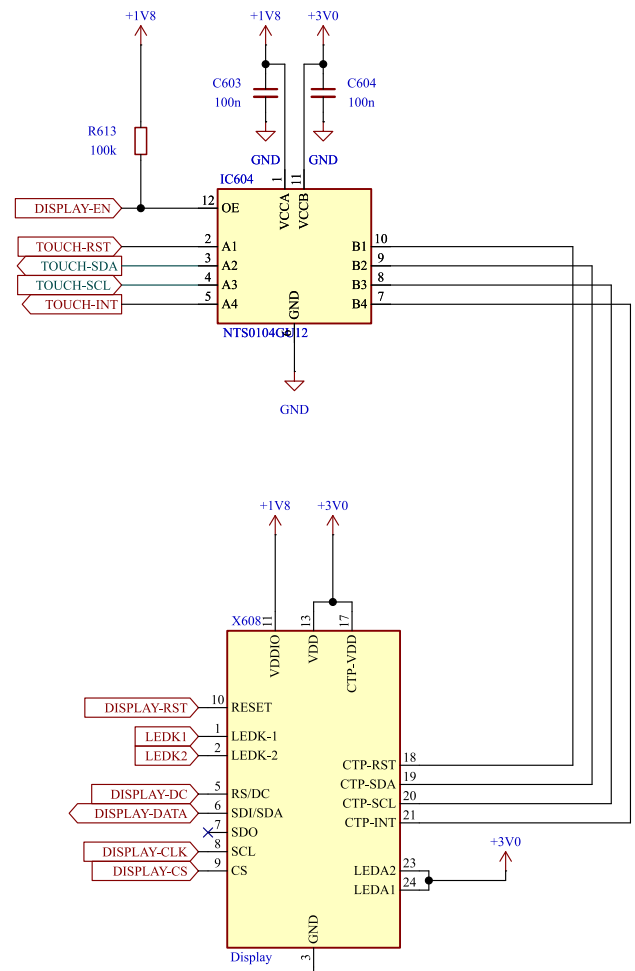


Microcontroller  
(NORA-B106/nRF5340)



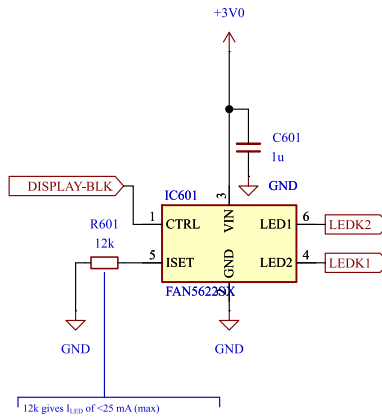
DESIGN NOTE:  
The SPI lines for the display can't be changed and are required for 32 MHz operation.

Display

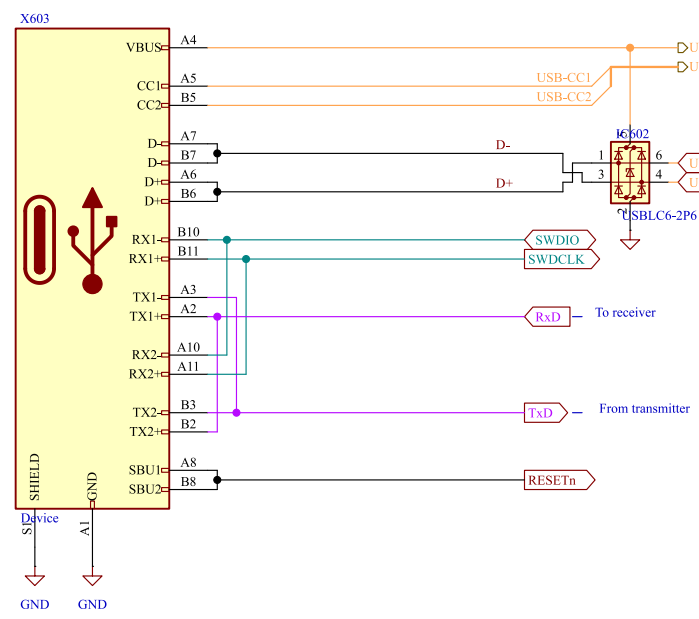


DESIGN NOTE:  
NTS0104GU12 has internal pullups.

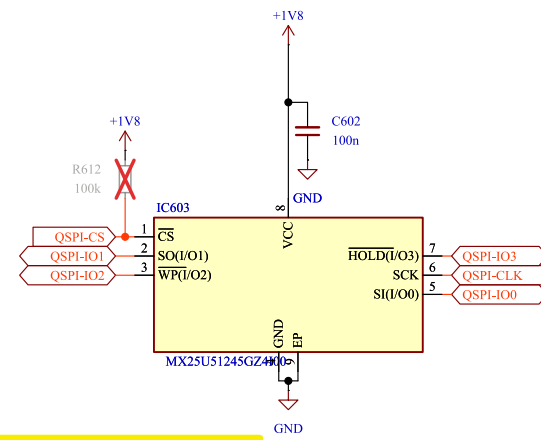
Display backlight



USB

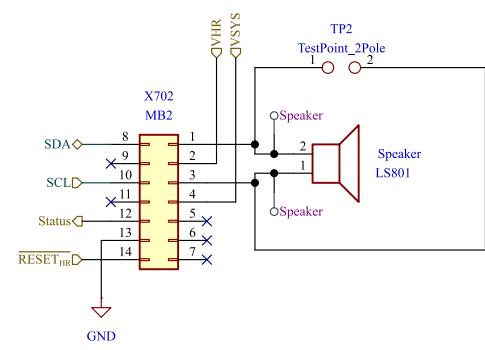
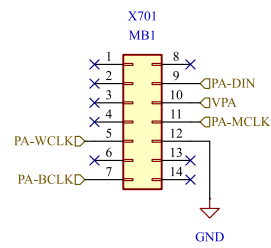


External Flash



DESIGN NOTE:  
Make sure that the Flash Memory supports 4k sector erase commands (limited by QSPI driver)!

|                                           |                                       |                             |                                              |                                  |                     |  |
|-------------------------------------------|---------------------------------------|-----------------------------|----------------------------------------------|----------------------------------|---------------------|--|
| Comments:                                 | Company:<br><b>ZSWatch</b>            |                             | Variant:<br>PRELIMINARY                      |                                  | Git Hash:<br>fcc54e |  |
|                                           | Board Name:<br><b>Watch-DevKit-HW</b> |                             | Project Name:<br><b>ZSWatch Watch-DevKit</b> |                                  |                     |  |
| Sheet Title:<br>MCU                       | File Name:<br>MCU.kicad_sch           | Designer:<br>Daniel Kampert | Date:<br>2025-07-13                          | Revision:<br>1.0.0+ (Unreleased) |                     |  |
| Sheet Path:<br>/Project Architecture/MCU/ |                                       | Reviewer:                   | Size:<br><b>A3</b>                           | Sheet:<br><b>6</b> of <b>9</b>   |                     |  |



|  |                                  |                     |                 |            |                      |           |
|--|----------------------------------|---------------------|-----------------|------------|----------------------|-----------|
|  | Comments:                        |                     | Company:        |            | Variant:             | Git Hash: |
|  |                                  |                     | ZSWatch         |            | PRELIMINARY          | fccc54e   |
|  |                                  |                     | Board Name:     |            | Project Name:        |           |
|  |                                  |                     | Watch-DevKit-HW |            | ZSWatch Watch-DevKit |           |
|  | Sheet Title:                     | File Name:          | Designer:       | Date:      | Revision:            |           |
|  | Extension                        | Extension.kicad_sch | Daniel Kampert  | 2025-07-13 | 1.0.0+ (Unreleased)  |           |
|  | Sheet Path:                      |                     | Reviewer:       | Size:      | Sheet:               |           |
|  | /Project Architecture/Extension/ |                     |                 | A4         | 7 of 9               |           |



# [8] Revision-History

**\${RELEASE\_TITLE\_1.0.0}**  
\${RELEASE\_BODY\_1.0.0}

**\${RELEASE\_TITLE\_...}**  
\${RELEASE\_BODY\_...}

**\${RELEASE\_TITLE\_...}**  
\${RELEASE\_BODY\_...}

**\${RELEASE\_TITLE\_...}**  
\${RELEASE\_BODY\_...}

|  |                                   |                                          |                             |                                       |                                  |
|--|-----------------------------------|------------------------------------------|-----------------------------|---------------------------------------|----------------------------------|
|  | Comments:                         | Company:<br>ZSWatch                      |                             | Variant:<br>PRELIMINARY               | Git Hash:<br>fccc54e             |
|  |                                   | Board Name:<br>Watch-DevKit-HW           |                             | Project Name:<br>ZSWatch Watch-DevKit |                                  |
|  | Sheet Title:<br>Revision-History  | File Name:<br>Revision History.kicad_sch | Designer:<br>Daniel Kampert | Date:<br>2025-07-13                   | Revision:<br>1.0.0+ (Unreleased) |
|  | Sheet Path:<br>/Revision History/ |                                          | Reviewer:                   | Size:<br>A4                           | Sheet:<br>8 of 9                 |